
Experience

Data Engineer – Amazon

August 2025 - Present

- Architect and lead development of Amazon LEO's first company-wide statistical data platform supporting ~20 cross-functional manufacturing and engineering teams, owning ~20 Spark/Glue pipelines processing ~120k+ manufacturing records daily
- Designed real-time production monitoring and alerting systems using AWS Lambda, DynamoDB, and Kinesis enabling automated detection of statistical anomalies saving 200+ engineering hours annually
- Engineered scalable ETL architecture performing complex transformations, multi-source joins, validation logic, and environment-based partitioning, improving dataset refresh performance by 50% (30 minutes to 15 minutes)
- Standardized pipeline architecture and deployment practices reducing analytics development cycles by >90% (1 month to 1-2 days)
- Develop predictive statistical models and AI models leveraging insight from the SPC platform to drive risk analysis and drive initial investigations, improving problem to resolution times by 40%
- Established automated data quality frameworks reducing defective/invalid records by 5% and improving downstream trust in operational reporting
- Serve as primary escalation point for VP-level and cross-functional quality investigations, delivering high impact root cause analyses and executive insights under tight timelines
- Co-designed and institutionalized SPC mechanisms, influencing technical and operational direction across engineering and production
- Mentored 3 engineering teams on SQL optimization, data modeling, and distributed systems best practices to elevate organizational engineering standards

Systems Engineer – Redwood Materials

August 2024 – August 2025

- Owned enterprise engineering systems including QMS, Jira, Tableau, Grafana, supporting quality and manufacturing operations
- Led redesign to the Incident Reporting and Problem-Solving application, reducing operation waste saving \$50,000+ annually
- Built internal engineering tools (Python, Java, React, FastAPI) improving lab, chemical, and materials workflows, saving \$35k+ annually
- Designed dashboards to track 2000+ samples, lab results, material flow, etc. for engineering departments saving \$20,000+ annually
- Automated manual tasks using Python and Power Automate, improving operational efficiency by 15%

Founder - Intsu Studios | Quality Engineering Technician – Redwood Materials | Quality Inspector – Panasonic Energy

Expertise

Industrial Data Engineering: Manufacturing Data Platforms, Data Standards, Predictive Modeling, Scalable Pipelines (batch/stream)

AWS Ecosystem: Glue, Redshift, S3, Lambda, Athena, Kinesis, DynamoDB, CloudWatch

Software Engineering: Python, Java, SQL, distributed systems, REST APIs, CI/CD

Education

Western Governors University – B.S Software Engineering

April 2023 – May 2025

Truckee Meadows Community College – General Education/Transfer

August 2019 – June 2021

Certifications

AWS Cloud Practitioner Certification

August 2024

ITIL V4 Foundations

August 2024

Google Project Management

August 2023

Google IT Support

June 2023